PACEE: Parent-Centered AI Scaffolding for Emotion Education in Early Childhood Conversations

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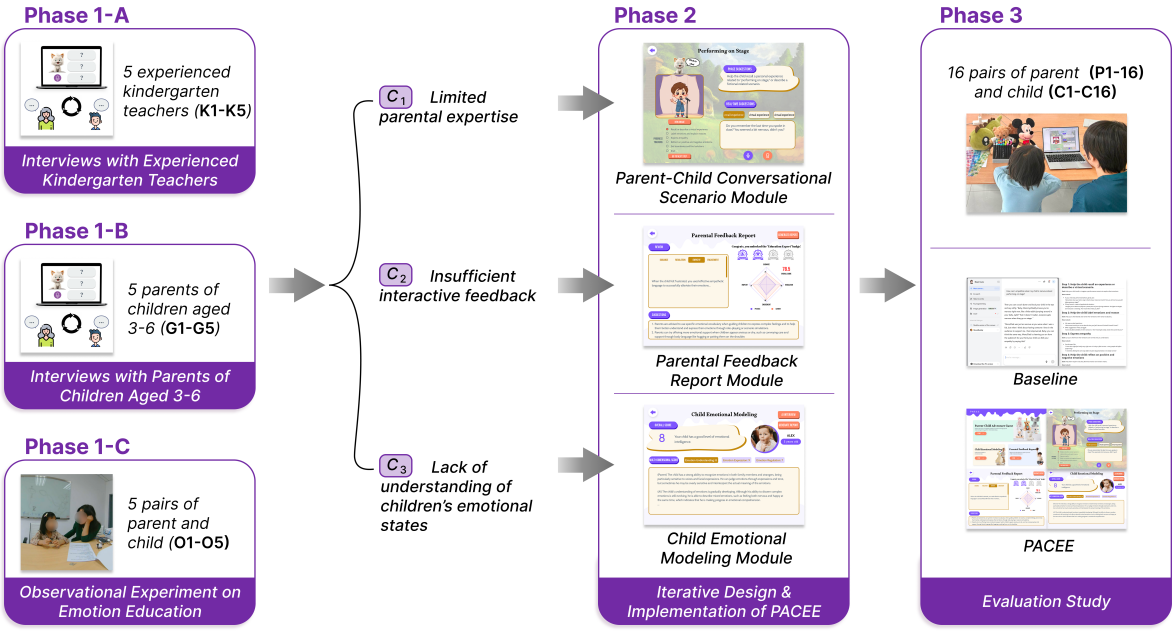


Figure 1: Our 3-phased study procedure.

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Abstract

Emotion education is critical for children aged 3 to 6. However, existing technologies depend substantially on direct child-AI interaction, overlooking the central role of parents in guiding early emotional development at home. To address this gap, we conducted co-design sessions with five kindergarten teachers and five parents to identify key parental challenges and opportunities for AI support in family emotion education. Based on these insights, we developed PACEE, an LLM-based assistant designed around an asymmetric triadic interaction framework: AI primarily supports parents while preserving parent-child interaction. PACEE provides parent-centered AI scaffolding that supports parents in real-time conversation through personalized guidance, post-hoc reflection through trackable feedback, and understanding children's emotional states through modeling. We evaluated PACEE with 16 families. Results showed that PACEE increased parent-child conversational engagement and improved parents' overall experience when guiding emotion-related conversations. Together, we contribute an asymmetric triadic interaction framework for AI-supported family education, offering design insights for AI systems that support parents while preserving parent-led parent–child interaction.

CCS Concepts

• **Human-centered computing → Collaborative interaction**.

Keywords

Parent-child Interaction, Emotional Education, Parent-AI Collaboration, LLMs

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1 INTRODUCTION

Emotion education is critical for children aged 3 to 6. During this period, children begin to understand emotions such as anger, fear, and sadness, and learn to express and regulate these feelings [11, 86, 110].

Recent work has explored AI systems that support children's emotional development, enabling them to recognize, express, and manage emotions. However, most child-AI dyadic systems and parent-child-AI triadic systems target children aged 8 and above, and rely substantially on direct interaction with AI [92, 95, 103]. Such child-facing designs are less suitable for younger children (ages 3–6) for two reasons: (1) young children experience emotions intensely but have limited ability to verbalize them [102], and (2) direct AI interaction in emotionally sensitive contexts may be inappropriate for young children, given their developmental vulnerability and limited ability to critically interpret AI responses [6, 76]. Moreover, prior child–AI systems often overlook parents' central role, while parent–child–AI triadic systems do not clearly weigh the importance of stakeholders. For children aged 3–6, we argue that parents should remain the primary guides because they mediate technology use and provide emotional security and co-regulation [4, 15, 32, 36, 78, 105]. Therefore, rather than designing AI to directly guide children or treating all three parties symmetrically, we reconceptualize triadic interaction by positioning parents as the primary stakeholder and propose an *asymmetric triadic parent–child–AI interaction framework* for AI-supported family emotion education.

However, positioning parents as the primary guides in this asymmetric triadic interaction also creates challenges. Many parents lack expertise in child psychology and communication strategies [31, 86, 87]. In practice, parents often struggle to interpret children's ambiguous emotional signals [20] and to support emotional regulation through conversation [77].

To better understand these parental challenges, we conducted interviews with five kindergarten teachers and five parents. Our findings reveal three key challenges: limited parental expertise, insufficient feedback, and a lack of understanding of children's emotional states. Based on these insights, we developed PACEE, an LLM-based assistant designed around asymmetric triadic interaction: it supports parents, rather than replacing them, in guiding emotion-related parent-child conversations. PACEE provides three modules, each addressing one identified challenge.

We evaluated PACEE's performance through a user study with 16 parent-child pairs. Compared to a baseline LLM-assisted approach, PACEE increased parent-child conversational engagement and improved parents' overall experience when guiding emotion-related conversations.

The contributions of this work are as follows:

- We propose an asymmetric triadic parent–child–AI interaction framework for AI-supported family education, where AI primarily supports parents while preserving parent-led parent–child interaction.
- We identify three key challenges in family emotion education through formative studies with teachers and parents, and derive corresponding design principles.
- We present PACEE, a parent-centered AI scaffolding system that supports parents in guiding children's emotional development during conversations.
- Through a user study with 16 families, we demonstrate the effectiveness of PACEE in enhancing parent-child emotion education, and provide design implications for parent-centered AI scaffolding in family education.

2 RELATED WORK

2.1 Emotion Coaching Techniques and Theories

Emotion coaching is a parenting technique that supports children's emotional understanding and regulation [40, 54]. A widely adopted framework in HCI is Gottman's Five Steps of Emotion Coaching [44, 45]: (1) Be aware of the child's emotion, (2) Recognize the child's expression of emotion as a perfect moment for intimacy and teaching, (3) Listen with empathy and validate the child's feelings, (4) Help the child learn to label their emotions with words, and (5) Set limits when helping the child to solve problems.

This guideline is often complemented by principles from positive psychology [8, 37], such as the PERMA model [91]. The PERMA model characterizes well-being through Positive Emotion, Engagement, Relationships, Meaning, and Achievement, and has been

applied in HCI areas such as well-being-centered design [64], persuasive technologies [79], and digital mental health [25].

Existing HCI systems that apply emotion coaching primarily target older children or clinical populations. Prior work explores AI-supported emotional expression and feedback for school-aged children and adolescents [59, 63, 92], as well as specialized support for conditions such as autism [42, 94, 103] or trauma [47]. However, little attention has been given to children aged 3–6.

Compared to older children, 3- to 6-year-olds experience emotions more situationally and intensely but have limited verbalization and self-regulation abilities [102]. These traits make parental guidance central to the development of emotional competence [46, 56]. Yet many parents face barriers, such as limited time and insufficient knowledge of emotion guidance [1, 16, 84]. Existing tools often assume children can interact with AI independently and fail to empower parents in their coaching role [93]. To address this gap, we propose a parent-centered, theory-grounded approach that supports parents as primary emotion coaches for children aged 3–6.

2.2 Parent-AI Collaborative Coaching System

Recent work has explored how AI supports children's emotional and cognitive development in family contexts. Existing systems generally fall into three categories based on AI's role in parent–child interaction: AI-centered, parent-centered, and parent–AI collaborative approaches.

AI-centered systems position AI as a teaching agent that interacts directly with children, providing personalized learning content [103], story-based learning [113, 117], and prompts to support language expression [92]. Parent-centered systems treat parents as primary educators, with AI as a support tool without direct child interaction. Prior studies have examined parents' needs and practices through interviews and co-design [28, 35, 58, 74, 81, 107], identifying opportunities such as media co-use [115] and reflective prompts for parents [59], though many stop at design insights with limited system-level support. Parent–AI collaborative systems combine both roles through task division: parents often act as facilitators or evaluators [22, 99, 114], while AI supports storytelling, communication, and structured activities [26, 61, 111, 119].

We extend the latter two lines of work to early childhood contexts by proposing a parent-centered, AI-supported coaching approach that strengthens parents' role in everyday emotion guidance.

2.3 Modeling Children's Emotional States

Personalized instruction is important for education and children's emotional development [12, 34]. Effective personalization requires modeling children's emotional states. Prior work addresses this through content frameworks and assessment methods.

For content design, the Test of Emotion Comprehension (TEC) [3, 83] is widely used to assess children's emotional competence with nine key elements, including emotion recognition emotion recognition [17], emotional expression [10], emotion external cause understanding [51], emotion-belief link [39], emotion reminder effect [50], emotional masking [41], mixed emotional understanding [7], emotional morality [53], and emotion regulation [5].

For assessment, traditional approaches rely on parent- or teacher-reported questionnaires [33, 85], while self-report tools such as TEIQue-CF [73] capture children's self-perceived competence. However, for children aged 3–6, limited cognitive and expressive abilities can reduce reliability [109]. More recently, AI-based systems using chatbots or games enable automated observation [80, 89], but often struggle to interpret subtle developmental cues [19].

To address these limitations, we propose a human–AI collaborative framework that combines parents' real-world understanding with AI-based analysis to support children's emotional modeling.

3 STUDY PROCEDURE

This study was approved by the Institutional Review Board, and we obtained consent from all participating parents and children. We conducted three study phases to explore, develop, and evaluate an AI assistant for supporting family emotion education (Figure 1).

Phase 1: Formative study. To understand needs and challenges in family emotion education, we conducted three formative studies: (1) interviews with 5 experienced kindergarten teachers (**K1–K5**), (2) interviews with 5 parents of children aged 3-6 (**G1–G5**), and (3) an observational experiment with 5 parent–child pairs (**O1–O5**). We identified three key challenges in family emotion education (C_1–C_3). Participants were recruited via social media, and each received a $10 stipend. Detailed demographics are provided in Appendix B.

Phase 2: System design and implementation. To mitigate these challenges, we developed three core modules of PACEE: Parent-Child Conversational Scenario (C_1), Parental Feedback Report (C_2), and Child Emotional Modeling (C_3). PACEE supports parents in operationalizing Gottman's Five Steps of Emotion Coaching [45] and incorporates design insights from the formative studies.

Phase 3: Evaluation study. We conducted a within-subject study with 16 parent–child pairs to evaluate PACEE, using a generic LLM chatbot as the baseline.

4 FORMATIVE STUDY

4.1 Procedure and Analysis

4.1.1 Need-finding Interview with Experienced Kindergarten Teachers. We interviewed 5 kindergarten teachers (3 female, 2 male; $M = 9.40$ years teaching experience, $SD = 6.34$) in one-hour video conference sessions. We discussed common negative emotions among children aged 3-6, their triggers, and teachers' guidance strategies, and explored how AI might support family emotion education. Using conceptual design images (Figure 2b), we co-designed potential AI-assisted interaction scenarios where parents guide children's emotions at home with AI support.

4.1.2 Need-finding Interview with Parents of Children Aged 3-6. Using the same conceptual design images, we interviewed 5 parents (4 mothers, 1 father; $M = 41.0$ years old, $SD = 3.16$) of children aged 3-6 in one-hour video conference sessions. Parents described their children's emotional behaviors, challenges in emotion guidance, and expectations for AI-assisted tools.

4.1.3 Observational Experiment on Children's Emotion Education. To understand parent–child interactions during emotion education, we recruited and observed 5 families (parents: 4 mothers, 1 father; $M = 38.2$ years old; children: 4 boys, 1 girl; $M = 4.8$ years old). In a 30-minute lab session, parents guided children through one of three negative emotion scenarios following Gottman's Five Steps

of Emotion Coaching [45]. Sessions were video-recorded, followed by short interviews about parents' experiences and difficulties.

All interview and observation data were transcribed and analyzed using thematic coding by three HCI researchers [30].



(a) (b)

Figure 2: Formative study scenes: (a) observational experiment setup; (b) conceptual design image used in interviews.

4.2 Findings

4.2.1 Key Challenges in Family Emotion Education. Through interviews with 5 kindergarten teachers (K1-K5) and 5 parents (G1-G5), we identified common scenarios where children aged 3–6 struggle with negative emotions that require repeated guidance. The findings serve as the initial entries for our system's scenario database (see Appendix B). Parental interviews and observational studies further revealed three key challenges.

C_1 **. Limited parental expertise.** All interviewed parents (5/5) reported lacking professional knowledge and practical strategies for guiding children's emotions. Parents struggled to translate general theoretical guidelines into concrete actions (G2: "*the examples in books don't always apply to my kid's situation*"), and observational data showed that they often "*don't know what to say*" in certain steps in the guideline (O1). Parents also emphasized the need for "*concrete, easy-to-implement language and techniques*" (G1, G2, G5).

C_2 **. Insufficient feedback.** Parents often lack objective feedback when guiding children's emotional development. As G3 noted, "*When he has strong emotions, I explain and comfort him in the moment, but I can't tell if he really understood or whether my approach was effective*". A similar uncertainty appeared in the observational study (O4), where parents followed the guidance steps but remained unsure whether the approach worked.

C_3 **. A lack of understanding of children's emotional states.** Parents reported difficulty accurately identifying children's emotions and their causes. This is partly due to limited visibility into children's school experiences (G4). Parents may misinterpret emotional triggers (G1) and also acknowledge subjective biases when evaluating their children's emotional development (G5), highlighting the need for more objective assessments.

4.2.2 Roles and Design Principles of AI in Family Emotion Education. Interviews with parents explored potential roles of AI in emotional guidance, while kindergarten teachers highlighted developmental characteristics of young children that should inform system design. All parents preferred to retain their primary role in emotional guidance and use AI to complement their limitations. They envisioned three main roles for AI.

First, AI should act as a **conversation facilitator**, dynamically adjusting its participation in parent–child interactions (G3: "*When parents take the lead, AI should speak less; when parents struggle, AI can step in to enrich the conversation.*"). Second, parents viewed AI as **an expert resource** that could integrate pedagogical knowledge into conversations and provide objective feedback about children's development. Third, AI could function as **a generative tool**, producing visual materials, conversation strategies, or performance feedback (G5: "*It could analyze which guidance techniques I applied successfully and how well I executed them.*").

Interviews with kindergarten teachers highlighted two design principles for AI-assisted emotion education systems. First, **systems should align with children's developmental stages**. Given young children's concrete thinking and limited language abilities, designs should incorporate visual aids and provide parents with age-appropriate communication guidance. Second, **systems should support personalized instruction**. Because emotional and cognitive development varies across children, AI systems should consider individual traits and behavioral patterns to deliver tailored guidance and developmental feedback.

5 DESIGN AND IMPLEMENTATION OF PACEE

5.1 User Interface and Design of System Modules

Based on insights from the formative studies, we designed PACEE, an LLM-based assistant that supports parents in guiding emotion-related conversations with their children. This design operationalizes an asymmetric triadic interaction framework: PACEE primarily supports parents and preserves parent-led parent-child interaction, with limited involvement of children in the AI interaction. The system contains three modules, each addressing a key challenge in family emotion education. Among them, the *Parental Feedback Report* and *Child Emotional Modeling* modules primarily support parents' understanding and reflection, while the *Parent-Child Conversational Scenario* module supports parent-led parent-child conversation.

5.1.1 Parent-Child Conversational Scenario. To address C_1 limited parental expertise, the *Parent-Child Conversational Scenario* module supports parents in conducting scenario-based conversations with their children. Parents guide conversations about their children's emotions, while PACEE provides real-time assistance. Parents begin by selecting a situation (e.g., performing on stage) from a scenario database derived from the formative study.

PACEE supports parents in operationalizing Gottman's Five Steps of Emotion Coaching [45] as a five-stage conversational workflow (Table 1). In this workflow, PACEE does not directly coach children; instead, it provides real-time support to parents across the five stages S_1-S_5. Specifically, PACEE facilitates conversation flow, provides stage-aware guidance, generates scenario images to help parents make situations more concrete, allows parents to optionally deliver AI-generated utterances when they need support in expressing their ideas more effectively, and tracks dialogue progress to support parents' awareness of the conversation. These functions primarily support parents in leading the parent-child conversation,

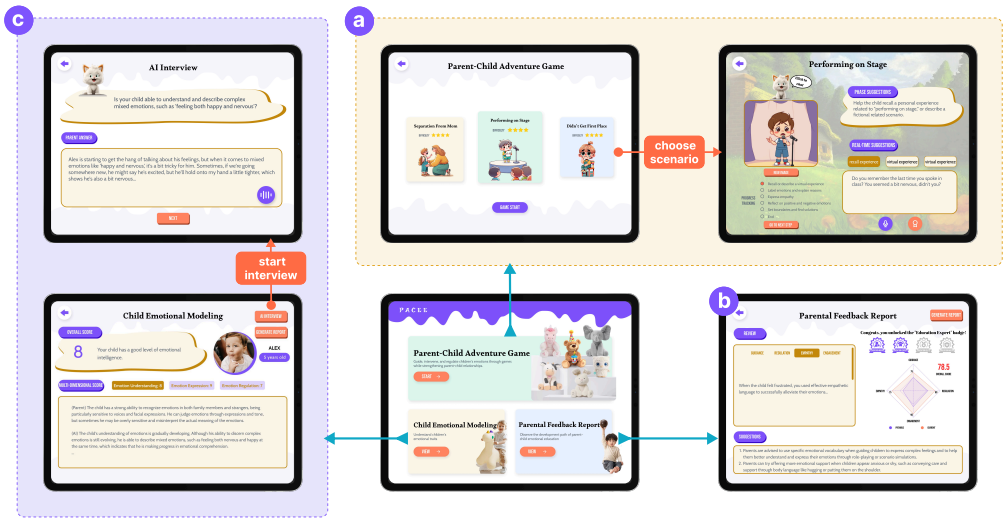


Figure 3: Main screens of PACEE showing its three modules: (a) Parent-Child Conversational Scenario, (b) Parental Feedback Report, and (c) Child Emotional Modeling.

while children's involvement is limited to viewing concrete scenario images and hearing AI-generated utterances when parents choose to use them.

Table 1: PACEE operationalizes Gottman's Five Steps of Emotion Coaching [45] to support parents interacting with children aged 3-6.

Tag	Gottman's Guideline [45]	Developmental Consideration of Children Aged 3-6	PACEE Support for Parents
S_1	Be aware of the child's emotion. Recognize the expression of emotions	Have limited memory recall. Need for concrete context.	Help the child recall an experience or describe a virtual scenario
S_2	Help the child learn to label their emotions	Need scaffolding to articulate feelings	Help the child label emotions and reason
S_3	Listen with empathy	Respond well to positive reinforcement (e.g., PERMA model [91])	Express empathy to children
S_4	Validate the child's feelings		Help the child reflect on positive and negative emotions
S_5	Set limits when helping the child to solve problem	Need help or collaboration in problem solving	Help the child set boundaries and find positive solutions

5.1.2 Parental Feedback Report. To address C_2 insufficient feedback, the *Parental Feedback Report* provides parent-facing feedback after *Parent-Child Conversational Scenario* sessions. It evaluates parental guidance during the parent-child conversations and provides actionable suggestions for future interactions.

PACEE analyzes transcripts of parent-child conversations and generates personalized recommendations grounded in established emotional education frameworks (e.g., Gottman's guideline [45]) and early childhood development guidelines for ages 3–6 [82]. The feedback evaluates how parents performed across the five stages of emotion coaching (S_1-S_5) and provides suggestions to improve their future guidance at each stage.

5.1.3 Child Emotional Modeling. To address C_3 limited understanding of children's emotional states, the *Child Emotional Modeling* module constructs external reference profiles for parental understanding and reflection, informed by the TEC framework [3, 83]. This module supports S_1 by helping parents recognize their children's emotional states and developmental characteristics, while also providing cues that facilitate empathetic listening in S_3. The model integrates two sources: (1) parents' long-term observations collected through a semi-structured AI interview, and (2) behavioral analysis of parent–child conversations during the *Parent-Child Conversational Scenario* (Figure 4).

A cognitive analyzer extracts updates from both sources across three dimensions (emotional understanding, emotional expression, and emotional regulation) and integrates them into the profile. The system also separates parent-reported observations and AI-driven analysis of conversation, making the source of each profile cue visible to parents. The resulting emotional profile is presented on the Child Emotional Modeling interface (Figure 3(c), helping parents reflect on their children's possible emotional signals to attend to in future conversational scenarios.

5.2 LLM-based Agent Workflow Design

We designed an LLM-based agent workflow for PACEE to support real-time parent–child interactions (Figure 5). During the *Parent-Child Conversational Scenario*, PACEE applies two complementary reasoning loops inspired by "thinking fast and slow" [27, 62, 71]: (1) rapid reasoning for parent-facing, real-time conversational support through low-latency generation and (2) reflective reasoning for modeling the child's emotional profile in the background.

For rapid reasoning, PACEE first transcribes parent-child conversations and uses parent-selected stages (S_1–S_5) to align interaction data. Within the selected stage, the conversation analyzer estimates real-time interaction progress while maintaining a structured representation of the conversation by organizing utterances and speaker turns. Based on the conversation context and the child's

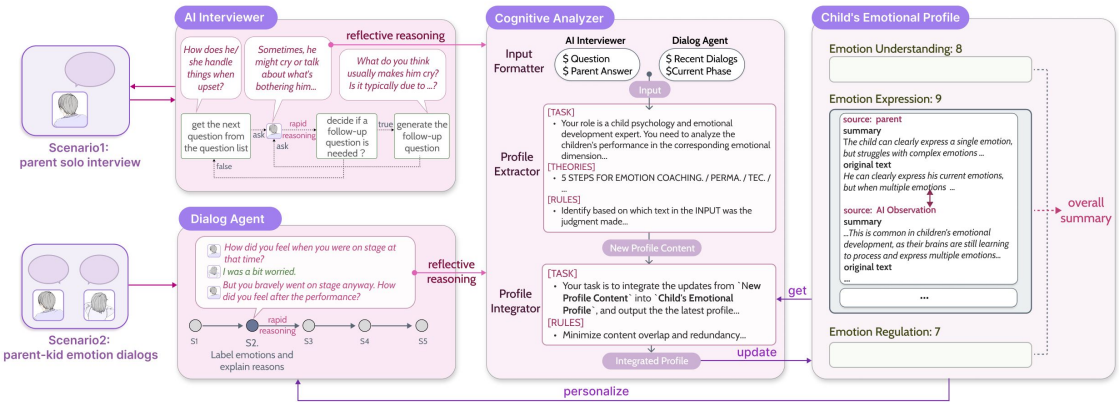


Figure 4: PACEE mechanism for collaboratively modeling children's emotional profiles through parent-AI interviews and parent–child conversations.

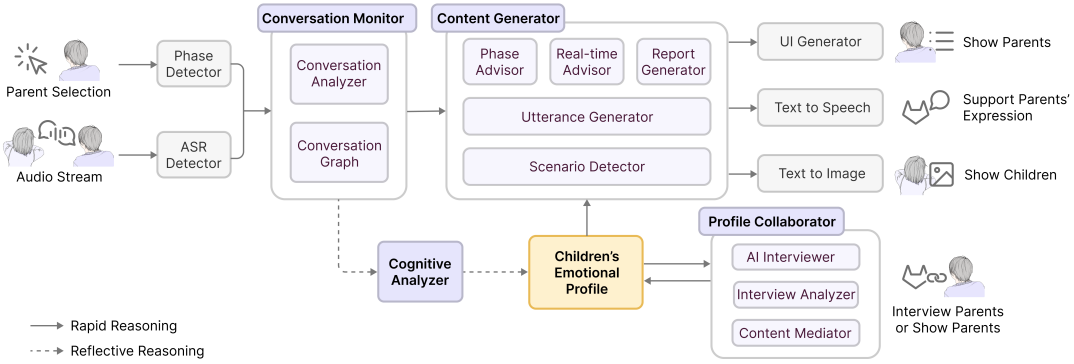


Figure 5: Agent workflow of PACEE.

emotional profile, the content generator applies rapid reasoning to provide real-time support, including stage-specific guidance, real-time suggestions, AI-generated utterances, and scenario images. In our study, we aimed to keep the content generator highly responsive, including real-time suggestions refreshed every 30 seconds, parent-triggered agent conversation, and parent-triggered image generation. To reduce latency, we selected lightweight LLMs with lower computational overhead and constrained LLM outputs to direct responses rather than lengthy reasoning traces.

In parallel, reflective reasoning is applied to update the child's emotional profile from conversation data. It serves as an asynchronous background process, so profile updates do not interrupt parent-AI real-time interaction. The cognitive analyzer periodically performs multi-step reasoning over accumulated dialogue to refine the profile during interactions (Section 5.1.3). To ensure quality, we used stronger reasoning models to process long dialogue data, extract and merge profile information, generating more complete reasoning traces during inference. The resulting profile content is labeled as AI-inferred conversational cues and kept separate from parent-reported observations.

Implementation details are provided in Appendix A.

6 USER EVALUATION

We conducted a within-subject study to evaluate PACEE in supporting family emotion education. The study aimed to answer the following questions:

RQ1. How does PACEE enhance family emotion education?

RQ2. How do parents collaborate with PACEE to develop children's emotional profiles and offer guidance to children?

RQ3. How do parents perceive PACEE in addressing their challenges in family emotion education?

We included a Baseline condition for comparison (Section 6.2). Each parent-child pair completed two emotional scenarios: one in the Baseline condition and the other in the PACEE condition, with scenario-condition assignments counterbalanced across participants.

6.1 Participants

We recruited 16 parent-child pairs through social media platforms. Parent participants (P1–16) included four fathers and twelve mothers (aged $M = 37.69$, $SD = 3.31$). Child participants (C1–16) included seven girls and nine boys aged 3–6 years ($M = 5.06$, $SD = 0.66$). Detailed demographics are provided in Appendix C.

The study followed standard ethical procedures. Parents provided informed consent for themselves and their children and could withdraw at any time. During the study, server-side moderation was conducted to review generated text and images, ensuring content safety for children.

6.2 Baseline Condition

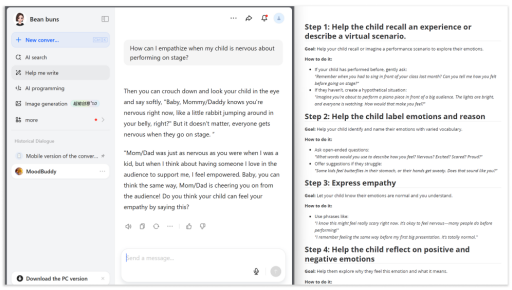


Figure 6: Baseline condition. Parents interacted with a voice-based LLM chatbot (Doubao) using the on-screen *Five Stages* guide, without PACEE features.

To evaluate PACEE at the whole-system level, we designed a Baseline in which parents interacted with a standard voice-based LLM chatbot (Doubao) through a web interface, while using an on-screen guide showing the *Five Stages* conversational workflow (Figure 6). Parents could request assistance from the chatbot freely during the interaction. This baseline represents an everyday setup in which parents use a general-purpose voice LLM with guideline materials, making it a meaningful comparison for PACEE.

We selected the Doubao chatbot [1] as the baseline for three reasons: (1) it is widely used among Chinese families, providing an ecologically valid comparison [2, 60, 116]; (2) it supports real-time Chinese voice interaction aligned with PACEE's hands-free workflow; and (3) it provides competitive performance on Chinese LLM benchmarks [112], ensuring the comparison focuses on interaction design rather than model capability.

The on-screen guide given to parents showed the same *Five Stages* workflow and core guideline principles as PACEE, ensuring structural consistency between conditions. While the stages were fixed, the specific conversational content remained open-ended.

6.3 Evaluation Design

We selected two scenarios from the system's database (Table 6), derived from the formative study: (1) *Not First*, where children may experience anger or disappointment after not winning first place, and (2) *On Stage*, where children may feel anxiety when performing in front of an audience.. We adopted a counterbalanced design. Each pair was randomly assigned to one of four groups in a 2×2 design, crossing the order of system conditions (Baseline vs. PACEE) and the assignment of emotional scenarios (*Not First* vs. *On Stage*).

[1] https://www.doubao.com/chat/

6.4 Procedure

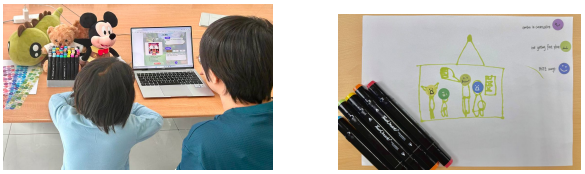
Each pair participated in a 2-hour lab study and interacted with the system on a touch-screen laptop (Figure 7).

Introduction. Parents received a brief system introduction and completed a short practice session. They were instructed to conduct the activity jointly with their child rather than letting the child interact with AI alone.

Parent-Child Emotional Conversation. Parents used PACEE or the Baseline to conduct emotional conversations with their children based on the assigned scenarios. To reduce potential discomfort for children aged 3–6, the researcher left the room and monitored the session remotely.

Post-Questionnaire. Parents completed a questionnaire including NASA-TLX [52], the Parenting Sense of Competence Scale (PSOC) [43], and items assessing how the systems addressed the three challenges (C_1–C_3), all rated on 7-point Likert scales.

Parent Interview and Child Drawing. We conducted semi-structured parent interviews, while children expressed their feelings using emoji stickers and drawings of the discussed scenarios.



(a) (b)

Figure 7: Scenes from the evaluation study. (a) Parent-child emotional conversation. (b) Child's drawing of the discussed scenario.

6.5 Data Analysis

For qualitative analysis, we transcribed all interviews and conducted thematic analysis [13, 14, 104]. Five HCI researchers collaboratively developed a codebook, independently coded the data, and resolved disagreements through discussion. Codes were then clustered to identify recurring themes.

For quantitative analysis, we transcribed the PACEE and Baseline session audio and performed automated text analysis. Emotion-related words were identified using the NTUSD dictionary [65]. We computed descriptive statistics and conducted paired-sample t-tests (interaction duration) and Wilcoxon signed-rank tests (dialogue turns, word counts, and emotion-related word counts).

7 FINDINGS

7.1 RQ1. How Does PACEE Enhance Family Emotion Education?

7.1.1 Increased Parent-Child Interaction. To compare observable parent-child conversation measures between PACEE and the Baseline, we analyzed interaction duration, dialogue turns, word count, and emotion-related word usage. PACEE showed higher values than the Baseline across these measures.

Interaction duration. Participants spent significantly longer time in sessions with PACEE (M = 892.37, SD = 310.96 vs. M =

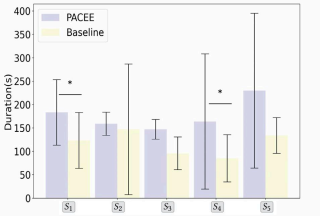


Figure 8: User interaction durations in each parent–child conversation stage. Significance results are marked with asterisks (*: p < .05).

560.17, $SD = 194.61$, $p < 0.01$). To further examine how interaction unfolded over time, we analyzed duration across the *Five Stages* (Figure 8). Participants using PACEE spent more time at each stage, with significant increases in S_1 and S_4 ($p < 0.05$ for both) .

Dialogue turns and word count. Participants engaged in more dialogue turns with PACEE ($M = 60$, $SD = 46.98$ vs. $M = 38$, $SD = 25.30$, $p < 0.05$). Parents produced more words ($M = 1597$, $SD = 678.56$ vs. $M = 994$, $SD = 468.36$, $p < 0.05$), and children showed a similar increase ($M = 370$, $SD = 240.43$ vs. $M = 165$, $SD = 102.25$, $p < 0.001$) .

Emotion-related word usage. Parents used more emotion words with PACEE ($M = 111$ vs. $M = 70$, $p < 0.05$), and children also showed increased emotional expression ($M = 19$ vs. $M = 10$, $p < 0.01$) .

7.1.2 Sustained Participation Over Time. To examine potential novelty effects, we divided each PACEE session into two equal temporal halves (Figure 9). No significant differences were found between the first and second halves in dialogue turns, word counts, or emotion-related words. This result reports within-session stability in these measures.

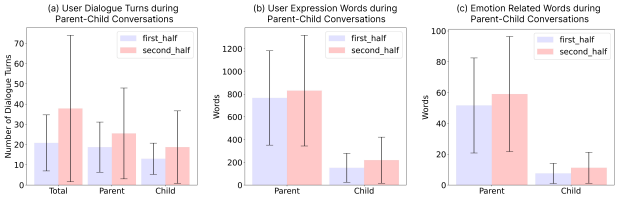


Figure 9: Temporal comparison of PACEE interactions between session halves: (a) dialogue turns, (b) word count, and (c) emotion-related words.

7.1.3 Improved Parental Experience and Children's Engagement Respectively. We analyzed NASA-TLX and PSOC results from the post-questionnaire. Compared to the Baseline, PACEE significantly improved parents' immediate interaction experience, including higher satisfaction ($p < 0.05, M = 5.06, SD = 1.12$ vs. $M = 3.94, SD = 1.57$), lower effort ($p < 0.01, M = 2.56, SD = 1.79$ vs. $M = 3.62, SD = 2.03$), and reduced frustration ($p < 0.05, M = 2.06, SD = 1.24$ vs. $M = 3.06$, $SD = 1.57$). In contrast, intrinsic psychological factors (e.g., self-efficacy) showed no significant changes, likely because they require longer-term interventions to produce measurable changes [9, 88].

We also analyzed children's post-study drawings to understand their perceptions of PACEE. 13 of the 16 children accurately reconstructed the interaction scenes (e.g., stage, parents, and audience) and frequently expressed positive emotions through visual cues such as smiles, hand-holding gestures, and warm color palettes. Additionally, most children (13/16) depicted PACEE as a friendly and supportive figure, often represented as smiling anthropomorphic characters or cartoon animals.

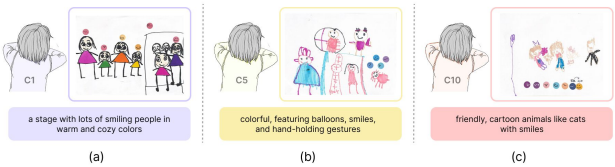


Figure 10: Drawings created by children (C1, C5, C10) after parent-child emotional conversations.

7.2 RQ2. How Do Parents Collaborate with PACEE to Develop Children's Emotional Profiles and Offer Guidance to Children?

7.2.1 Parents and PACEE Contribute Different Factors in Construction of Children's Emotional Profiles. We annotated the system interaction logs using the method described in Section 6.5. As shown in Table 2, the results indicate that parents and PACEE play complementary roles in constructing children's emotional profiles. PACEE primarily derives insights from parent-child interaction data, whereas parents contribute contextual knowledge from firsthand observations, third-party feedback, school progress, and comparisons with how the child behaved at younger ages. In interpreting children's emotions, parents tend to rely on everyday life events and experiential knowledge of behavioral habits, whereas PACEE analyzes interaction patterns to assess children's emotional and developmental characteristics.

7.2.2 PACEE Provides New Insights on Children's Profile for Parental Reflection. Parents reported that PACEE offered structured external interpretations of children's emotional profiles, revealing perspectives they had previously overlooked. Some realized their earlier judgments were either underestimated or overly optimistic. For example, P2 noted that the AI revealed emotional understanding he had not recognized in his child, while P4 considered the AI's evaluation more rational than her own judgment.

PACEE also helped parents reflect on causes of children's emotional reactions. As P8 explained, "*He was very reluctant to go on stage. I assumed he was nervous...but the AI suggested that the anxiety might have stemmed from the absence of his mother's support.*"

Overall, most parents perceived these discrepancies between parents and AI as valuable reflections. 14 of 16 parents found PACEE's analysis to be a useful external perspective that complemented their experience. However, two parents questioned the reliability of some interpretations, noting that AI might misread subtle emotions or fail to detect when children are not telling the truth.

Table 2: Parents and PACEE contribute different themes in children's emotional profiles in the evaluation studies.

Source	Theme	Sub-theme (Partial)	Example
Parent	Knowledge and Events	Emotions learned at school; Proficiency in describing certain emotions	C2: *Complex emotions, such as fatigue, are typically guided appropriately by kindergarten teachers in class, so they should be distinguishable.*
	Contextual Responses	Emotional reactions or regulation strategies for different scenarios	C11: *When he fails in a game, it tends to trigger his specific feelings of frustration.*
	Behavioral Habits	Habits of mood adjustment; Seeking external help; Tendency to go to parents	C6: *At present, she cannot calmly face negative emotions; she still seeks help from his mother or those she feels close to.*
	Growth Trends	Developmental changes from earlier ages	C12: *When he is happy, he directly expresses his joy. When he was younger, he would jump around and even scream and shout.*
PACEE	Specific Emotional Capacity	Understanding, expressing, or positively regulating specific emotions	C15: *When facing possible failure or competition, she considers responding to challenges through effort and learning, showing her progress in stressful situations.*
	Macro Emotional Capacity	Ability to identify emotion origins or others' emotions	C2: *...and can recognize the emergence of these emotions in specific situations (such as speaking on stage). He can feel the impact of the audience's reactions on himself. This indicates he can identify the source and influence of emotions.*
	Capacity Development Trends	Emotional capacities being developed	C14: *He demonstrates a certain awareness of strategies for emotional regulation, indicating he is developing the ability to regulate emotions and can attempt to use different methods to cope with emotions.*
	Languaguistic Characteristics	Directness of language expression; Use of non-verbal communication methods	C13: *He can directly express his emotional state but may lack a more extensive and nuanced vocabulary to describe his feelings.*

7.2.3 Parents Appreciate Open and Positive Advice from PACEE. During the study, parents received a total of 513 real-time advice updates from PACEE. Log and video analysis showed that parents most frequently adopted suggestions related to collaborative problem-solving and open-ended questioning (Appendix C).

Parents also mentioned the usefulness of open-ended questioning in the interview, as P2 stated, "*Real-time advice helps me realize some issues I hadn't thought about, such as prompting my child to provide more details. Sometimes I think it's okay when a child says one word, but the advice reminds me to ask for more details.*"

Other types of advice, including scenario simulation, concrete suggestions, positive encouragement, and empathy and acceptance, were also adopted by some parents.

7.2.4 Parents Accept PACEE in Collaborative Guidance, but with Concerns. Parents generally accepted PACEE as a collaborative assistant in parent–child emotional guidance, while emphasizing that parents should remain the primary decision-makers. Most participants reported that they contributed more to the guidance process, with PACEE mainly providing professional suggestions that parents could selectively adopt.

All parents welcomed PACEE's involvement, though preferred interaction modes varied. Some envisioned PACEE interacting directly with children as a peer-like educator, while others preferred it to support parents only or act as a neutral mediator.

However, parents emphasized clear boundaries for AI involvement. Some rejected features that impose goals on children (P4: "*Setting goals for children... I think it is up to parents to do it.*"), or intervene in sensitive family matters (P1: "*Don't press further... for example, on topics that parents find taboo and children find sensitive.*"). Parents also stressed that PACEE should not replace emotional intimacy in parent–child relationships.

7.3 RQ3. How Do Parents Perceive PACEE in Addressing Their Challenges in Family Emotion Education?

Parents' ratings show that PACEE significantly outperformed the Baseline in addressing the three key challenges of family emotion education (each $p < 0.01$; Figure 11). Interviews further indicate that parents perceived PACEE as playing multiple supportive roles in mitigating these challenges.

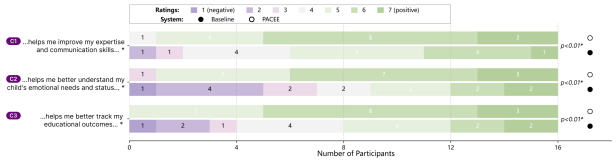


Figure 11: Post-questionnaire results on how PACEE and Baseline address the three key challenges.

7.3.1 Enhancing Parental Expertise and Communication Confidence. Regarding C_1 limited parental expertise, the *Parent-Child Conversational Scenario* module supported parents who reported not knowing "what to say" or "how to guide" children. Parents experienced difficulties across several steps: recalling experiences ($n = 4$), labeling emotions ($n = 8$), expressing empathy ($n = 8$), reflecting on emotions ($n = 5$), and finding positive solutions ($n = 6$).

PACEE's suggestions provided structured guidance across these stages. For example, P8 noted that the step-by-step process enabled smoother interactions, while others reported that the suggestions helped them adopt the child's perspective and steer conversations.

The module also increased parents' confidence: 10 participants reported that the suggestions offered concrete, child-friendly ways to express emotions, helping them organize thoughts and communicate more openly. Most parents reported greater initiative and comfort in emotional conversations. However, some parents noted that overreliance on AI-generated expressions might weaken their own communication abilities.

7.3.2 Providing Trackable Feedback and Actionable Future Strategies. Addressing C_2 insufficient feedback, the *Parental Feedback Report* module mitigated issues such as "lack of timely feedback" and "invisible outcomes". Eight parents reported that AI-generated analyses revealed previously unnoticed shortcomings, prompting self-reflection (e.g., P6: "*The system reminded me to avoid interrupting. I hadn't realized I was doing that.*"). Five parents also valued structured summaries of conversation highlights, which made parenting efforts more visible and trackable.

Beyond reviewing past interactions, the module provided actionable strategies for future communication. Parents appreciated the evidence-based suggestions and practical perspectives for handling children's emotions. However, some concerns remained, including the limited ability of a single session to reveal long-term outcomes and the need for more personalized suggestions.

7.3.3 Improving Parental Reflection on Children's Developmental Trajectories. Regarding C_3, a lack of understanding of children's emotional states, the *Child Emotional Modeling* module provided

data-driven cues for parents to reflect on children's emotional traits and behavioral patterns. Twelve parents reported that the system introduced "new knowledge and perspectives," helping them overcome cognitive blind spots and reflect more objectively on their children's emotions (e.g., P12: "*The system's questions prompted me to think about aspects I hadn't paid attention to before. The conclusions helped me understand my child more systematically and comprehensively.*")

Parents also discussed the alignment between their views and the system's analyses: some reported high consistency that strengthened trust, while others found discrepancies useful for reconsidering their perceptions. Several parents further valued that the system contextualized their child's behavior within broader developmental patterns, supporting parental reflection on age-related characteristics and future development.

8 DISCUSSION

In this work, we introduced PACEE, an AI assistant to support parents in family emotion education. We found that PACEE can support their reflection and improve their satisfaction when guiding children's emotions. We clarify that our study does not claim that PACEE improves parents' emotional coaching skills. Instead, it structures parent-child conversations following established emotional coaching steps, helping parents remain engaged and supporting their reflection. Therefore, the improved conversational engagement should not be interpreted as evidence of validated emotional coaching effectiveness, but as a socio-technical system to guide parent-child conversation. Assessing parents' emotional coaching skills would require expert validation, which is beyond the scope of this study and could be pursued in future work.

Here, we further discuss the impact of PACEE on Gottman's Five Steps of Emotion Coaching [45]. We then propose broader design implications for parent-centered, AI-supported education systems.

8.1 Parent-AI Co-construction of Children's Emotional Understanding

A central challenge in parent-centered AI systems is helping parents better understand their children. Our findings suggest that this understanding should emerge through parent-AI collaboration rather than solely through AI-driven assessment.

Prior systems typically rely either on parent-reported information [23, 38, 119] or AI-based assessment [66]. In contrast, PACEE integrates both: parents contribute experiential knowledge from everyday observations, while AI analyzes parent–child interaction data to identify emotional and developmental characteristics. Therefore, in family contexts, AI may be most effective not as an authority diagnosing children, but as a reflective partner that augments parents' understanding. Future parent-centered AI systems should therefore support the co-construction of child models with parents, combining experiential knowledge with data-driven analysis.

8.2 Rethinking Gottman's Five Steps as a Triadic Parent-Child-AI Collaboration

Gottman's model assumes a dyadic parent-child interaction in which children can recall, articulate, and reflect on emotions [45] and has been applied in AI-supported HCI contexts, especially in AI-supported emotional education with schoolchildren in a dyadic format [59, 92, 103]. However, this dyad might not fit well for preschool children aged 3-6 who are still developing their emotional responses to challenges, and their parents also face challenges in sustaining developmentally appropriate guidance at home. Thus, we adapted Gottman's Five Steps of Emotion Coaching and propose a triadic *parent-child-AI* collaboration mode in which AI collaborates with parents to guide their children's emotional development.

AI can help parents support emotion recognition by externalizing children's vague or fragmented experiences. Because preschoolers often struggle to recall and verbalize the causes of emotions [21, 72], AI-generated scenarios can make emotional situations more concrete. For young children, emotion recognition should be understood less as eliciting verbal explanations and more as helping parents and children jointly interpret emotions through concrete scenarios.

Empathy in AI-supported family education can be understood as a shared interpretive process rather than solely an individual parental skill. Parents often find empathetic listening difficult to sustain in daily interactions. PACEE's *Child Emotional Modeling* module provides an additional perspective on children's emotional states, complementing parents' contextual knowledge with structured developmental cues. This can reduce subjective blind spots and facilitate deeper emotional understanding between parents and children. More broadly, our findings extend empathy from a purely interpersonal skill to a shared interpretive process in which AI supports parents' perspective-taking while parents still lead the emotional relationship.

Support for young children's problem solving should prioritize positive-resource scaffolding over solution prescription. Prior work shows that children aged 3-6 respond well to positive reinforcement [49, 97]. Building on this insight, PACEE foregrounds positive emotions and strengths through the PERMA model [91]. In early childhood contexts, the goal of problem-solving should not be narrowly defined as correcting negative behavior or prescribing fixed solutions, but rather as helping children gradually build emotional resources for future coping.

8.3 Asymmetric Triadic Interaction for AI-supported Family Education

Prior UIST work on AI-supported family and educational systems has largely focused on enabling direct child–AI interaction, positioning children as independent users [18, 24, 48, 69, 96, 118, 120]. A smaller line of work introduces AI into parent-child interaction [106], but still assumes that children actively engage with AI.

In contrast, we introduce an *asymmetric triadic parent–child–AI interaction framework*, where AI primarily supports parents while preserving parent-led parent–child interaction. The asymmetry lies not only in who receives AI outputs, but also in how responsibility is distributed: AI provides scaffolds, parents interpret and decide whether to use them, and children participate mainly through the parent-child relationship. This reframes AI from child-facing to parent-augmenting, reflecting the constraints of sensitive contexts—especially for young children—where direct child-AI interaction may be developmentally inappropriate or risky.

This asymmetric framework can be extended to settings where children face challenges in directly interacting with AI and rely on caregivers as intermediaries. It extends prior work on supporting children with autism spectrum disorder [57, 68, 108], cognitive impairments [67], and language delays [100, 101]. Beyond early childhood education, it may also inform other sensitive caregiving contexts in which children need adult mediation, such as moral guidance, safety-critical instruction, health-related routines, and family-privacy-sensitive conversations. In these contexts, AI should strengthen caregivers' ability to guide and reflect, rather than becoming an additional authority that children must negotiate directly.

More broadly, our findings suggest the need to rethink multiparty human–AI interaction beyond symmetric role assumptions. In high-stakes family and educational contexts, rather than engaging all participants equally, *who AI interacts with—and to what extent—* becomes a critical design decision. Designing for asymmetric participation, where AI selectively supports certain roles while remaining minimally intrusive to others, may be essential for ensuring safety and appropriateness.

At the same time, applying generative AI in such sensitive contexts raises important challenges. Because PACEE's guidance is AI-generated and lacks expert validation, incorrect or poorly contextualized suggestions may interfere with effective emotional coaching. Therefore, caregiver control and expert-informed safeguards are central to this interaction framework, and systems like PACEE should be positioned as assistive tools that augment caregivers' reflection rather than replace professional expertise.

Taken together, PACEE serves as a design probe exploring how generative AI may support parent-led education in everyday family contexts. Future systems should integrate insights from developmental psychology and incorporate expert-informed safeguards.

8.4 Design Considerations for Parent-centered, AI-supported Education Systems

Parent interviews and prior work suggest three design considerations for *parent-centered, AI-supported education systems.*

Customize AI support for parents as well as children. While prior work primarily adapts AI coaching to children's profiles [29, 103, 119], our findings show that parents also vary in their expectations regarding the difficulty, timing, and frequency of AI assistance. Future systems should therefore adapt AI support to both children's characteristics and parental preferences.

Enable transparent and negotiable parent-AI collaboration. Parents expressed concerns about the reliability of AI interpretations and potential disagreement with AI assessments. Prior work suggests that transparency improves human–AI collaboration [70]. Systems should therefore explain how suggestions are generated and allow parents to question or adjust AI interpretations.

Respect parental boundaries and preserve parental autonomy. Parents consistently viewed themselves as primary decision-makers in their children's education and were cautious about AI overstepping family boundaries. Consistent with prior work on parental decision-making [98], our findings extend this insight to emotion-sensitive contexts in early childhood education. AI systems should avoid overly prescriptive guidance and instead support flexible, parent-controlled use.

8.5 Limitations and Future Work

This study has several limitations. First, participants were limited to Chinese families, and most parent participants were mothers, reflecting common caregiving patterns for children aged 3–6 [55, 75, 90]. Future work should examine broader cultural contexts and include more balanced parental roles. Second, our design cannot fully isolate the foundation model variable. While our focus is the interaction framework rather than raw model output (Section 6.2), future work could adopt a 2×2 factorial design–(Interface: Baseline vs. PACEE) $\times$ (Model: Doubao vs. GPT-4o)–to separate interface and model effects. Future work should also compare PACEE with intermediate baselines, such as versions without real-time guidance, feedback reports, or child profile support, to isolate the contribution of each module. Third, although both conditions used the same Five Stages workflow, duration, dialogue turns, and word-based measures should still be interpreted with caution: they may reflect PACEE's active scaffolding and mirroring of AI-suggested language rather than engagement alone. Future work could separate original parent-child utterances from system-suggested phrases, or analyze how parents adapt, reject, and personalize AI suggestions during conversation. Fourth, lower NASA-TLX effort or frustration likely reflects PACEE reducing parents' moment-to-moment uncertainty through offloading conversational planning burden, rather than long-term skill improvement. Longitudinal studies are needed to examine whether repeated use builds parents' emotion-coaching skills or creates dependency. Fifth, PACEE's child emotional profile should not be interpreted as a validated psychological assessment. In practice, further safeguards are needed to reduce the risk of parental overtrust or misleading inferences. Validating such profiles would require expert assessment and longitudinal evidence. Finally, the study was conducted in a lab setting using a touchscreen laptop. Future work should conduct longitudinal in-home deployments to examine parent–AI collaboration in everyday family environments.

9 CONCLUSION

We presented PACEE, an LLM-based assistant that helps parents guide their children's emotional development. In a study with 16 parent–child pairs, PACEE increased parent-child conversational engagement and improved parents' overall experience in guiding emotion-related conversations. changeATogether, PACEE builds toward an asymmetric triadic interaction framework for AI-supported family education. We hope this work advances parent-centered family education and inspires AI systems that support caregivers while preserving sensitive parent-child interactions.

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Table 5: Demographic details of participants in the observational experiments.

Alias	Parent		Child	
	Age	Gender	Age	Gender
O1	37	Male	3	Boy
O2	37	Female	6	Boy
O3	41	Female	4	Boy
O4	43	Female	6	Boy
O5	33	Female	5	Girl

A PACEE System Implementation

We implemented the core agent workflow in Python on a Flask server. The PACEE interface, written in Vue, interacts with the server by exchanging messages via WebSocket. Children's emotional profile and conversation graph information are stored in separate JSON files.

For text generation tasks that were less latency-sensitive, we used the GPT-4o API, which offered relatively strong output quality at the time of our study. To minimize latency in parent-facing conversational support, the faster GPT-4o-mini model was adopted. Speech-to-text processing utilized the TINGWU API, with audio sliced and compressed on the frontend before transcription. Text-to-speech conversion relied on Alibaba Cloud's streaming synthesis API, while image generation integrated the Stable Diffusion XL (SDXL) model.

For moderation, a researcher monitored server-side generated text and images in real time during the user study. If child-inappropriate or emotionally unsafe content was detected, the researcher was instructed to prevent it from being displayed to the family. Parents also remained the final mediators of whether to use AI-generated suggestions with children.

B Additional Formative Study Details

This section provides supplementary details about the formative study, including (1) participant demographics (Table 3-Table 5) and (2) the scenario database identified from the study, which captures common situations where children aged 3-6 experience negative emotions and require repeated parental guidance (Table 6).

Table 3: Demographic details of kindergarten teachers in the need-finding interviews.

Alias	Age	Gender
K1	31	Male
K2	29	Female
K3	30	Male
K4	48	Female
K5	31	Female

Table 4: Demographic details of parents in the need-finding interviews.

Alias	Age	Gender
G1	37	Female
G2	45	Female
G3	41	Female
G4	38	Male
G5	44	Female

C Additional User Evaluation Details

This section provides supplementary details about the user evaluation, including (1) participant demographics (Table 7) and (2) an analysis of real-time advice accepted by parents during the study (Table 8).

Table 6: Our need-finding interviews identified common scenarios of negative emotions experienced by children aged 3-6.

Type	Example Scenarios	Common Emotions
Separation	Leaving parents for school; missing parents; waking up alone	Anxiety, sadness
Peer conflict	Disputes over toys/turns; sibling quarrels; having belongings taken	Anger, frustration
Social setbacks	Being ignored by peers; unable to join new groups	Loneliness, sadness
Physical discomfort	Injury, illness, falls, or grogginess upon waking	Distress, sadness
Autonomy violation	Forced task termination (e.g., iPad taken away); not winning first place	Anger, disappointment
Negative feedback	Teacher corrections; critical remarks from peers	Sadness, shame
Stressful challenges	Challenging sports/games; performing on stage	Anxiety, fear

Table 7: Participant demographic information in the user evaluation.

Child ID	Child Gender	Child Age	Parent ID	Parent Gender	Parent Age
C1	Female	5	P1	Male	41
C2	Male	5	P2	Male	36
C3	Female	4	P3	Female	37
C4	Male	5	P4	Female	43
C5	Female	5	P5	Female	38
C6	Female	4	P6	Female	43
C7	Female	5	P7	Female	34
C8	Male	6	P8	Female	36
C9	Female	5	P9	Female	39
C10	Female	4	P10	Female	29
C11	Male	5	P11	Female	39
C12	Male	5	P12	Female	39
C13	Male	6	P13	Male	39
C14	Male	5	P14	Female	36
C15	Male	6	P15	Male	36
C16	Male	6	P16	Female	38

Table 8: Categories of real-Time advice accepted by parents during the user evaluation.

Type	Number	Description	Example
Open-ended Questioning	22	Asking open-ended questions to encourage children to describe their emotions, thoughts, or experiences.	*Do you have any special feelings when redeeming your points? Do you feel happy or a bit unsure?*
Scenario Simulation	16	Setting up or recalling specific scenarios to encourage role-playing or imagination.	*Imagine yourself as a brave little hero, like My Little Pony—you can show your courage and talent on stage.*
Concrete Suggestions	7	Providing clear, specific strategies to help children manage emotions or enhance positive feelings.	*We can go to the small playground on the rooftop for a run, or maybe find something delicious to eat.*
Positive Encouragement	9	Using praise and positive feedback to reinforce children's positive behaviors and emotional experiences.	*You did a great job jumping rope—210 jumps is really impressive!*
Empathy and Acceptance	16	Expressing empathy and acceptance towards children's emotions, making them feel understood and supported.	*I understand you might feel nervous on such a big stage. Dad has felt like that before, too.*
Collaborative Problem-solving	31	Discussing solutions together with children.	*Let's think together about what we can do to perform even better in the next competition.*